

Long-run logistics-based control of non-immunizing infectious diseases

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ARTICLE INFO

Article history:

Received 10 November 2025

Received in revised form 26 January 2026

Accepted 25 March 2026

Available online 27 March 2026

Handling Editor: Dr Yijun Lou

Keywords:

Non-immunizing infectious diseases

Long-term planning

Optimal control

Stability analysis

SIS epidemic-inventory dynamic model

ABSTRACT

Non-immunizing infectious diseases cannot be eradicated and therefore generate persistent health and intervention costs over time. Steering such diseases toward a stable endemic equilibrium is a natural long-run policy objective. This paper contributes by formulating an integrated infinite-horizon SIS-inventory control model to capture the spread of non-immunizing infectious diseases. Inventory is governed by a controllable replenishment rate that influences the recovery rate, directly linking logistical capacity to epidemiological outcomes. We derive regime-dependent trajectories implied by the Pontryagin Maximum Principle and establish conditions for the existence, reachability, and saddle-type local stability of steady states under three tractable pure replenishment regimes: zero, moderate, and maximum. We find that zero replenishment fails to support convergence to the disease-free equilibrium and leads to persistence of infection, whereas maximum replenishment yields a unique endemic steady state. In contrast, moderate replenishment may generate multiple equilibria. Threshold effects emerge, showing that insufficient replenishment can undermine system stabilization. Numerical experiments illustrate transition dynamics, convergence speeds, and sensitivity to key parameters, yielding practical insights for long-run epidemic management.

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1. Background

Non-immunizing infectious diseases continue to be a significant health threat worldwide. These diseases do not confer lasting immunity, which allows infections to persist over time. The Susceptible-Infected-Susceptible (SIS) model is commonly used to describe such dynamics (Elahi et al., 2025). In SIS models, susceptible individuals who become infected through contact with infectious individuals eventually recover and return to the susceptible class. Practical applications of SIS epidemic models include meningococcal disease (Gruhn et al., 2024) and gonorrhoea (Kline et al., 2025).

Controlled SIS models have been extensively studied as tools for mitigating the spread. Several studies consider finite-horizon control problems. For instance, Gajardo et al. (2021) examined screening policies in a prison population to control a communicable disease, while Williams et al. (2024) compared vaccination and screening regimes for gonorrhoea control. These works provide valuable insights into short-term intervention design. However, they are not intended to address long-run control policies.

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Peer review under the responsibility of KeAi Communications Co., Ltd.

To capture the persistent nature of non-immunizing diseases, researchers have employed infinite-horizon control formulations. Specifically, Rowthorn et al. (2009) analyzed optimal treatment strategies over an infinite horizon across two interconnected subpopulations. They showed that treatment should be prioritized in the region with the lower infection rate. Herrmann and Nkuiya (2017) studied optimal substitution strategies between two antibiotics with different resistance profiles. Chehrizi et al. (2019) incorporated drug quality dynamics and showed that the optimal control has a bang-bang structure with a single switching time point. More recently, Delmas et al. (2025) developed a theoretical framework for targeted vaccination in an infinite-dimensional SIS setting. While these studies focus on optimal policy design over time, they do not explore the long-run behaviour of the disease, including equilibrium characterization, threshold conditions, and steady-state convergence.

A growing body of literature explicitly investigates the long-run behaviour of controlled SIS models. La Torre et al. (2020) studied long-run intervention regimes and transitional dynamics. Goenka and Liu (2020a) studied the role of infectious diseases in a growth framework. They showed that the optimal public health subsidy can be lower in economies trapped in poverty than in growing economies. Rowthorn and Toxvaerd (2020) analyzed optimal prevention and treatment and found that, in a steady state, prevention is interior while treatment is either zero or maximal. Bosi et al. (2021) analyzed the impact of altruistic behavior under a lockdown control policy and found that the optimal intensity of lockdown depends on the degree of altruism. Recent contributions have added to and extended this line of research by incorporating behavioral responses (Parino et al., 2024), network effects and risk aversion (Bizyaeva et al., 2024), static intervention policy ensuring disease elimination (Zino et al., 2025), and an epidemic game-theoretic approach (Maitra et al., 2025).

Despite this progress, an important dimension remains almost underexplored. Specifically, existing SIS infinite-horizon control models largely overlook logistics-related measures (e.g., inventory management). In practice, these factors are vital for ensuring timely and effective treatment of infected individuals, and neglecting logistics constraints in epidemiological models may lead to suboptimal managerial decisions, potentially compromising population health.

Several studies have attempted to incorporate resource constraints into disease dynamics. For instance, Zhao et al. (2020) and Bugalia et al. (2021) embedded resource limitations into epidemic models through parameters affecting treatment or recovery rates. Wang (2006) and Wang et al. (2019) introduced piecewise-defined functions linking treatment rates to the availability of medical resources. More recently, Gandzha et al. (2021), Liu et al. (2023), and Zhang et al. (2025) modeled the evolution of resource stocks using a dynamic inventory equation coupled with SIS epidemic dynamics. These studies analyzed the stability of the resulting systems. However, they do not treat inventory replenishment as a control and therefore do not constitute a controlled SIS-inventory model over an infinite horizon.

The present study aims to fill this gap. We develop an infinite-horizon control model that integrates SIS epidemic dynamics with a dynamic inventory equation. The replenishment rate is treated as a control variable, and it influences the recovery rate, directly linking logistical capacity to epidemiological outcomes. However, we acknowledge that, in real-world applications, frequent adjustments to replenishment policies are likely to be impractical, due to administrative, logistical, or financial constraints. Consequently, this study focuses on the long-run behavior of the system under pure control strategies—specifically, zero, moderate, and maximum replenishment rates—while it excludes mixed strategies. Investigating mixed regimes could offer a promising direction for future research.

The resulting problem is analyzed using Pontryagin's Maximum Principle. We determine the steady states, characterize their stability, and explore the transition dynamics to the steady states under various control regimes. This analysis reveals how inventory levels, captured through policy replenishment regimes, can affect long-run epidemic outcomes, including equilibrium infection levels and convergence behavior. The insights derived may support decision-makers in designing effective long-term policies for combating non-immunizing infectious diseases.

The main contributions of this paper are as follows:

- To the best of our knowledge, this study is the first to formulate an infinite-horizon optimal control model that explicitly couples SIS epidemic dynamics with a controlled inventory system to study the long-run behavior of the disease.
- We characterize the steady states and stability properties of the resulting coupled SIS-inventory system under three pure inventory replenishment regimes.
- We analyze how alternative inventory control policies affect stabilization time and the transition dynamics toward the steady states.

The remainder of the paper is structured as follows. The mathematical model is described in Section 2, followed by numerical computations and analysis in Section 3. Finally, Section 4 concludes the paper with a discussion of the findings and suggestions for future research.

2. Mathematical model

We begin by introducing the main parameters and notations used throughout this paper (see Table 1). Note that Table 1 is divided into two parts. The first part presents the parameters used in the model, along with their corresponding description, baseline values, and units. The second part includes the state, co-state, and other time-dependent variables (e.g., a control variable), for which only the corresponding description is provided.

In our problem setting, a non-immunizing infectious disease spreads within a closed population of fixed size N . This population is divided into two classes: susceptible individuals (S) and infected individuals (I). Infected individuals transmit the disease to susceptibles via direct contact at a rate β (transmission rate). It is assumed that all infected individuals can

Table 1

Notations.

Notation	Description	Value [Units]
<i>Parameters:</i>		
N	Total population (assumed constant over time): $N = S(t) + I(t)$	1000 [individuals]
β	Transmission rate	$0.3 \left[\frac{1}{\text{time unit}} \right]$
γ	Baseline recovery rate (the recovery rate in the absence of any medication)	$0.1 \left[\frac{1}{\text{time unit}} \right]$
z	Maximum quantity of medication delivered per replenishment cycle	$10 \left[\frac{\text{units}}{\text{time unit}} \right]$
δ	Medication consumption rate per infected individual	$1.5 \left[\frac{\text{units per infected person}}{\text{time unit}} \right]$
ρ	Discount rate	0.05
α	Scaling parameter that determines how strongly the available stock influences the recovery rate	0.001
$u^{\max}$	Maximal replenishment rate	$8 \left[\frac{\text{replenishments}}{\text{time unit}} \right]$
a_1	Cost paid per infected individual per unit time	$\$40 \left[\frac{\text{infected person}}{\text{time unit}} \right]$
a_2	Holding cost per unit of medicine in stock per unit time	$\$10 \left[\frac{\text{unit in stock}}{\text{time unit}} \right]$
a_3	Cost associated with the replenishment effort u per unit time	$\$30 \left[\frac{\text{replenishment}}{\text{time unit}} \right]$
<i>State, co-state, and other time-dependent variables:</i>		
$S(t)$	Size of susceptible population at time t	
$I(t)$	Size of infected population at time t	
$Q(t)$	Quantity of available medication at time t	
$u(t)$	Stock replenishment rate (control variable) at time t . It indicates the intensity or frequency of replenishment efforts	
$\varphi_I(t), \varphi_Q(t)$	Co-state variables at time t	
$\tilde{\gamma}(t)$	Actual recovery rate at time t	
F	Objective function (measured in \$)	

recover at a rate $\tilde{\gamma}$. Recovered individuals then re-enter the susceptible class and may become reinfected. The disease transmission dynamics just described can be captured by the SIS epidemic model (see, for example, [Hethcote, 1976](#)), as represented by equations (1) and (2).

$$\dot{S}(t) = -\frac{\beta S(t)I(t)}{N} + \tilde{\gamma}(t)I(t), S(0) = S^0, \quad (1)$$

$$\dot{I}(t) = \frac{\beta S(t)I(t)}{N} - \tilde{\gamma}(t)I(t), I(0) = I^0. \quad (2)$$

Given that $N = S(t) + I(t)$ (see [Table 1](#)), the dynamics of system (1)–(2) reduce to

$$\dot{I}(t) = \frac{\beta(N - I(t))I(t)}{N} - \tilde{\gamma}(t)I(t), I(0) = I^0. \quad (3)$$

In addition to the disease dynamics, we introduce a dynamic equation to describe the evolution of the medication inventory:

$$\dot{Q}(t) = zu(t) - \delta I(t), Q(0) = Q^0. \quad (4)$$

Equation (4) indicates that the inventory level $Q(t)$ increases with the replenishment effort (a control variable) $u(t)$, scaled by the delivery capacity z , while it decreases due to consumption at a rate δ . We assume that only infected individuals consume the medication (e.g., antibiotics); thus, the total amount of consumed medication at any given time point is $\delta I(t)$ (see (4)).

When medication is readily available (that is, when $Q(t)$ is high), infected individuals receive treatment quickly, leading to faster recovery and a lower probability of further transmission. In such cases, the recovery rate $\tilde{\gamma}(t)$ increases. Conversely, when medication is scarce, infected individuals may experience delays in receiving treatment, leading to slower recovery and higher disease prevalence. Therefore, the recovery rate is assumed to be positively associated with the inventory level. This aligns with the works of [Paul and Venkateswaran \(2016\)](#) and [Alqahtani et al. \(2025\)](#). Specifically, [Paul and Venkateswaran \(2016\)](#) reported that patients who receive treatment have a shorter average recovery time than those who do not. However, establishing a precise relationship between the available medication stock and the recovery rate is challenging. In the current work, we follow a common modeling approach in epidemic optimal-control studies (see [Grandits](#)

et al., 2019; Mahadhika & Aldila, 2024; Martcheva, 2015). In these studies, the recovery rate is linearly related to the sum of a baseline recovery rate and a time-dependent control variable reflecting treatment intensity. In our coupled epidemic-inventory setting, treatment availability is not exhibited through a control variable directly acting on infected individuals but instead via the medication stock $Q(t)$. Hence, we define the recovery rate as $\tilde{\gamma}(t) = \gamma + \alpha Q(t)$, where γ is the baseline recovery rate in the absence of medication and α is a scaling parameter capturing the degree to which the available stock enhances recovery. In other words, α reflects the effectiveness of medical stock in improving health outcomes: higher values of α denote a more substantial influence of $Q(t)$ on $\tilde{\gamma}(t)$. The value of this parameter is disease- and drug-specific, since the impact of medication on the course of a disease can vary significantly depending on the disease and the treatment. We note that a linear relationship between the recovery rate and the medicine inventory is adopted as a simplifying approximation; in practice, additional medication yields smaller recovery gains once healthcare capacity limits are reached.

Next, substituting $\tilde{\gamma}(t) = \gamma + \alpha Q(t)$ into (3) yields

$$\dot{I}(t) = \frac{\beta(N - I(t))I(t)}{N} - (\gamma + \alpha Q(t))I(t), I(0) = I^0. \quad (5)$$

From (5), it follows that if $\gamma > \beta$, i.e., if the baseline recovery rate exceeds the disease transmission rate, then the number of infected individuals will naturally decline continuously, eventually reaching the disease-free equilibrium. In other words, $I(t \rightarrow \infty) = 0$, regardless of the inventory dynamics, as time tends to infinity. However, since the primary purpose of the current study is to investigate the impact of the available stock of medication on the disease dynamics, we restrict our attention to cases where $\gamma < \beta$.

The state variables in (4)–(5) are naturally non-negative, i.e., $I(t) \geq 0$ and $Q(t) \geq 0$ for $t \in [0, \infty]$. The non-negativity of $I(t)$ can be demonstrated by solving the corresponding differential equation (5) with a known, non-negative initial value I^0 , which leads to

$$I(t) = \frac{e^{\int_0^t (-\alpha Q + \beta - \gamma) d\tau}}{\int_0^t \left[\frac{e^{\int_0^\tau (-\alpha Q + \beta - \gamma) d\tau}}{N} \beta \right] d\tau + I^0} \geq 0 \quad (5.1)$$

Unlike the state variable $I(t)$, which is naturally non-negative (see (5.1)), the state variable $Q(t)$, defined by (4) and representing the available medicine stock, may fall below zero. To prevent this, we require that

$$Q(t) \geq 0, \text{ for } t \in [0, \infty]. \quad (6)$$

Furthermore, the control variable $u(t)$, representing the stock replenishment effort, is bounded. Thus, we require that

$$0 \leq u(t) \leq u^{\max}, \quad (7)$$

where $u^{\max}$ denotes the maximum feasible replenishment effort, which is the maximum number of times at which stock can be replenished during a predefined period. This upper bound depends on the logistical capacity of the healthcare system. For example, a larger vehicle fleet or a more efficient distribution infrastructure would enable more frequent stock replenishment.

The goal of the mathematical model defined by (4)–(7) is to determine the optimal replenishment rate that minimizes the total cost incurred from the following three sources: (a) the presence of infected individuals within the population, (b) inventory holding costs, and (c) stock replenishment efforts – all evaluated over an infinite planning horizon. In mathematical terms, the objective function can be defined as

$$F = \int_0^\infty e^{-\rho t} \left[\frac{a_1 I(t)^2}{2} + \frac{a_2 Q(t)^2}{2} + \frac{a_3 u(t)^2}{2} \right] dt \rightarrow \min. \quad (8)$$

To conduct a regime-wise steady-state analysis, we proceed according to the following road map: First, we define the Hamiltonian function (see (9)). Using the Hamiltonian, we derive the canonical system, which consists of the state and co-state differential equations (see (4)–(5) and (10)–(11), respectively). Depending on the structure of the optimal control, the system may admit either bang-bang or interior regimes. By substituting each regime into the canonical system and setting the time derivatives to zero, we obtain the corresponding steady states, which are then analyzed with respect to their feasibility and stability (see Subsections 2.1–2.3). In our analysis, we assume that constraint (6) is never binding; thus, the corresponding co-state variable does not exhibit jumps. Additionally, for notational simplicity, we omit the time index t when the temporal dependence is evident from the context.

Following the above road map, and based on equations (4) and (5) and the objective function (8), the Hamiltonian function is defined as

$$H = -\frac{a_1 I^2}{2} - \frac{a_2 Q^2}{2} - \frac{a_3 u^2}{2} + \varphi_I \left(\frac{\beta(N-I)I}{N} - (\gamma + \alpha Q)I \right) + \varphi_Q (zu - \delta I), \quad (9)$$

and the corresponding co-state equations are

$$\dot{\varphi}_I = \varphi_I \rho - \frac{\partial H}{\partial I} = \varphi_I \rho + \frac{((\alpha Q - \beta + \gamma)\varphi_I + a_1 I + \delta \varphi_Q)N + 2\beta I \varphi_I}{N}, \quad (10)$$

$$\dot{\varphi}_Q = \varphi_Q \rho - \frac{\partial H}{\partial Q} = \varphi_Q \rho + \alpha \varphi_I I + a_2 Q, \quad (11)$$

subject to the following transversality conditions:

$$\lim_{t \rightarrow \infty} e^{-\rho t} \varphi_I(t) I(t) = 0 \text{ and } \lim_{t \rightarrow \infty} e^{-\rho t} \varphi_Q(t) Q(t) = 0. \quad (12)$$

Equations (4), (5) and (10)–(11) together form the canonical system of the problem under investigation. Since the Hamiltonian is quadratic in the control variable u (see (9)), and considering the bound stipulated in (7), the optimal regime may be either of the bang-bang form, where $u = 0$ or $u = u^{\max}$, or of the interior form. The interior solution, if it exists, can be derived from (9) and is given by $u = \frac{z\varphi_Q}{a_3}$. For each regime of the control variable, we now investigate the steady states and the corresponding optimal transition paths, i.e., the trajectories that connect the initial values of the state and co-state variables to their respective steady states. We begin with the case of a zero-replenishment policy, i.e., $u = 0$ (see Sub-section 2.1).

Before exploring the steady state under various regimes, we first clarify the meaning of the *admissible steady states* that will be used throughout the paper. For the state variables, i.e., $I(t)$ and $Q(t)$, the corresponding steady states, denoted by I^{ss} and Q^{ss} , are admissible if $I^{ss}, Q^{ss} \geq 0$. Specifically, when $I^{ss} = 0$, the steady state is referred to as a *disease-free state*, whereas $I^{ss} > 0$ indicates an *endemic steady state*. In contrast, the interpretation of the steady states for the co-state variables, denoted by φ_I^{ss} and φ_Q^{ss} , differs. Specifically, following the epidemic-control literature (see Forster & Gilligan, 2007; Goenka & Liu, 2020a), we focus on the regime in which the co-state variable associated with the infected population is negative at steady state, i.e., $\varphi_I^{ss} < 0$. In particular, Forster and Gilligan (2007) explained that the co-state variable related to the infected population reflects the marginal benefit of reducing the stock of infection and, therefore, is typically negative. In turn, the co-state variable φ_Q^{ss} associated with the medicine stock at the steady-state may have any sign.

2.1. Zero-replenishment policy ($u = 0$)

The zero-replenishment policy arises when $u = \frac{z\varphi_Q}{a_3} < 0$, which occurs when $\varphi_Q < 0$. In this case, the marginal effect of increasing the medicine stock is negative, meaning that stock replenishment is no longer worthwhile. As a result, it is optimal to cease replenishment efforts. In the following proposition, we show that under the zero-replenishment policy (i.e., when $u = 0$), the system does not admit a unique steady state.

Proposition 1. *Under the zero-replenishment policy, the steady state is given by*

$$I^{ss} = 0, Q^{ss} = Q, \varphi_I^{ss} = \frac{Qa_2\delta}{\rho(\alpha Q - \beta + \gamma + \rho)}, \varphi_Q^{ss} = -\frac{Qa_2}{\rho} \quad (13)$$

Proof: Substituting $u = 0$ into equations (4), (5) and (10)–(11), and setting the derivatives equal to zero, yields (13). ■

As can be seen from (13), the steady state under the zero-replenishment regime corresponds to a disease-free equilibrium. However, the medical stock level Q^{ss} can take on different values at the steady state, meaning that the equilibrium in (13) is not unique.

Proposition 2. *The steady state in (13) is a non-hyperbolic saddle point with a local one-dimensional stable manifold.*

Proof: Calculating the eigenvalues of the Jacobian matrix evaluated at the steady state yields

$$\lambda_1 = 0, \lambda_2 = -\alpha Q + \beta - \gamma, \lambda_3 = \rho, \lambda_4 = \alpha Q - \beta + \gamma + \rho. \quad (14)$$

The zero eigenvalue $\lambda_1 = 0$ indicates that the steady state is non-hyperbolic. Since $\lambda_3 = \rho > 0$, the steady state possesses at least one unstable direction. To prove that among the remaining eigenvalues, i.e., λ_2 and λ_4 , only the eigenvalue λ_4 is negative, we consider the sign of the co-state variable φ_I associated with infected individuals at the steady state, i.e., φ_I^{ss} . Since we focus on the regime in which $\varphi_I^{ss} < 0$, as commonly assumed in the epidemic-control literature, from (13) it follows that $\varphi_I^{ss} < 0$ if and only if $\alpha Q - \beta + \gamma + \rho < 0$ (evaluated at $Q = Q^{ss}$). Under this condition, one can easily verify that λ_4 in (14) is negative while λ_2 is positive. Hence, the steady state is a non-hyperbolic saddle point with a local one-dimensional stable manifold. ■

We now proceed to show that any admissible trajectory cannot reach this non-hyperbolic saddle point (13).

Proposition 3. *The steady state (13) is not reachable by any admissible trajectory.*

Proof: To prove the statement in the proposition, we show that $I^{ss} = 0$ in (13) is not reachable as $t \rightarrow \infty$. The proof proceeds by contradiction.

Recall equation (5), which may be rewritten as follows:

$$\dot{I}(t) = I(t) \left[\beta - \gamma - \alpha Q(t) - \frac{\beta I(t)}{N} \right], I(0) = I^0. \quad (15)$$

Assume that the steady state $(I^{ss}, Q^{ss}) = (0, Q)$ is reached as $t \rightarrow \infty$, i.e., $I(t) \rightarrow 0$ and $Q(t) \rightarrow Q$ as $t \rightarrow \infty$. Since $\beta - \gamma - \alpha Q > \rho > 0$ (see Proposition 2), define

$$\eta := \beta - \gamma - \alpha Q - \rho > 0,$$

so that $\beta - \gamma - \alpha Q = \eta + \rho$. Let $f(q) = \beta - \gamma - \alpha q$. Since $Q(t) \rightarrow Q$ as $t \rightarrow \infty$ and $f(q)$ is continuous, it follows that

$$f(Q(t)) \rightarrow f(Q) = \beta - \gamma - \alpha Q = \eta + \rho$$

Then, by the definition of the limit, there exists a time $\check{t}_1 > 0$ such that for all $t \geq \check{t}_1$

$$|f(Q(t)) - (\eta + \rho)| < \varepsilon_1,$$

where $\varepsilon_1 > 0$. Let $\varepsilon_1 := \frac{\eta}{2}$, which implies

$$\beta - \gamma - \alpha Q(t) \geq \rho + \frac{\eta}{2}$$

Next, assume that $I(t) \rightarrow 0$ as $t \rightarrow \infty$. Then, by the definition of the limit, there exists a time $\check{t}_2$ such that $0 < I(t) < \varepsilon_2$ for all $t \geq \check{t}_2 > 0$,

where $\varepsilon_2 > 0$. Let $\varepsilon_2 = \frac{(\rho + \frac{\eta}{2})N}{2\beta} > 0$ and define $\check{t} := \max \{\check{t}_1, \check{t}_2\}$. Then, for all $t \geq \check{t}$, we have

$$\beta - \gamma - \alpha Q(t) - \frac{\beta I(t)}{N} \geq \rho + \frac{\eta}{2} - \frac{\beta I(t)}{N} \geq \rho + \frac{\eta}{2} - \frac{\beta \varepsilon_2}{N} = \frac{\rho + \frac{\eta}{2}}{2} > 0$$

which, when substituted into (15), leads to

$$\dot{I}(t) = I(t) \left[\beta - \gamma - \alpha Q(t) - \frac{\beta I(t)}{N} \right] > \left(\frac{\rho + \frac{\eta}{2}}{2} \right) I(t) \text{ for all } t \geq \check{t}.$$

Hence, $I(t)$ increases for all $t \geq \check{t}$, which contradicts the assumption that $I(t) \rightarrow 0$ as $t \rightarrow \infty$. ■

The zero replenishment regime represents a scenario in which the healthcare system halts medication supply. This situation may arise when available logistics resources must be redirected to more urgent missions (e.g., mitigation of natural or man-made disasters) or due to budgetary constraints that force decision-makers to reconsider priorities. Mathematically, as shown in Proposition 3, this policy renders the disease-free equilibrium unreachable: in the absence of the medication inflow, the disease cannot be eradicated. From a policy perspective, this implies that a zero-replenishment effort cannot serve as a stand-alone long-term strategy for disease control. Instead, it should be complemented by other interventions, such as screening, early detection programs, and public awareness campaigns, to prevent surges in infection levels.

Next, we analyze the steady states and the optimal transition paths under the maximum replenishment policy.

2.2. Maximum replenishment policy ($u = u^{max}$)

The maximum replenishment policy arises when $u = \frac{Z\varphi_Q}{a_3} > u^{max}$, which occurs when $\varphi_Q > \frac{a_3 u^{max}}{Z}$. Expressed in words, this means that it is beneficial to apply the maximum possible replenishment effort only when the marginal benefit of increasing the medicine stock exceeds the product of the marginal cost of deploying the maximal logistics effort and the maximal replenishment rate divided by the delivery capacity.

In the following proposition, we show that the steady state under $u = u^{max}$ is unique.

Proposition 4. The steady state is unique and is given by

$$\begin{aligned} I^{ss} &= \frac{u^{max} Z}{\delta}, Q^{ss} = \frac{-N(\gamma - \beta)\delta - u^{max} Z \beta}{N\alpha\delta}, \varphi_I^{ss} = \frac{Na_2(\gamma - \beta)\delta^2 + a_2\beta u^{max} Z \delta + Na_1\alpha\rho u^{max} Z}{\alpha(N(\alpha u^{max} Z - \rho^2)\delta - \beta\rho u^{max} Z)}, \varphi_Q^{ss} \\ &= \frac{(-\rho a_2(\gamma - \beta)\delta^2 - (u^{max} Z \alpha)^2 a_1)N^2 - u^{max} Z a_2 \beta \delta(\gamma + \rho - \beta)N - a_2(\beta u^{max} Z)^2}{N\delta\alpha(N(\alpha u^{max} Z - \rho^2)\delta - \beta\rho u^{max} Z)} \end{aligned} \quad (16)$$

Proof: Substituting $u = u^{\max}$ into equations (4), (5) and (10)–(11), and setting the derivatives equal to zero, yields (16). ■

The steady-state value of the infected population, $I^{ss} = \frac{u^{\max}z}{\delta}$, indicates that, in the long run, the disease reaches an endemic state. In this state, the disease may persist at a constant level in the population, without causing new outbreaks, provided the replenishment efforts remain at the maximum possible level. Furthermore, as the maximum replenishment effort ($u^{\max}$) or the maximum delivery capacity per replenishment (z) increases, the steady-state number of infected individuals rises. From a policy perspective, I^{ss} represents a capacity-constrained steady state; a greater supply capacity, i.e., $u^{\max}z$, enables the system to sustain a larger pool of the infected population without losing stability. However, maintaining the supply capacity at the same level while increasing the medication consumption rate δ , reflecting the intensity of treatment per infected person, may diminish that capacity effect, thereby leading to a reduced I^{ss} .

As previously stated, we focus on the case where $\gamma < \beta$. Relaxing this condition would result in an infeasible steady-state value for the amount of available medical stock, Q^{ss} , as can be seen from (16). This condition implies that maintaining a medicine stock in the long run is only justified when the natural recovery rate (γ) is lower than the rate of disease transmission (β). From the policy standpoint, this means that for the healthcare system, maintaining medical stock is justified only when the disease cannot be naturally eliminated from the population, and exogenous efforts are required. If, on the other hand, $\gamma > \beta$, then the disease will naturally die out over time without external intervention, making continuous stockpiling of medicine unnecessary.

Furthermore, the steady-state expression for the medicine stock (Q^{ss}) exhibits different operational behaviour compared to I^{ss} . Unlike I^{ss} , where a greater supply capacity ($u^{\max}z$) has a positive effect on the amount of the infected population that can be sustained in the steady state, $u^{\max}z$ has an adverse effect on Q^{ss} . From a logistical perspective, this means that greater healthcare logistics agility, enabled by maximum replenishment capacity, reduces the necessity for large physical stockpiles in the long run.

In the following Corollary 1, we define the necessary conditions to ensure that the steady states of I^{ss} , Q^{ss} and φ_I^{ss} in (16) are admissible, i.e., $I^{ss}, Q^{ss} \geq 0$ and $\varphi_I^{ss} \leq 0$.

Corollary 1. The steady states defined in (16) are admissible (i.e., $I^{ss}, Q^{ss} \geq 0$ and $\varphi_I^{ss} \leq 0$), provided that the following conditions hold

$$\alpha N \delta - \beta \rho > 0 \text{ (i) and } \frac{N \delta \rho^2}{z(\alpha N \delta - \beta \rho)} < u^{\max} \leq \min \left\{ -\frac{N(\gamma - \beta)\delta}{z\beta}, -\frac{Na_2(\gamma - \beta)\delta^2}{z(\alpha Na_1\rho + a_2\beta\delta)} \right\} \text{ (ii).} \quad (17)$$

Proof: The result follows directly from the steady-state expressions in (16) and the conditions specified above, and is therefore omitted. ■

We now outline, in Proposition 5, the stability of the steady state defined in equation (16). To facilitate the analysis, let J denote the Jacobian matrix derived by linearizing the canonical system of equations (4), (5) and (10)–(11), evaluated under the policy $u = u^{\max}$. Additionally, let Φ be the magnitude of J , defined as the sum of its second-order principal minors minus the squared discount rate.

Proposition 5. Suppose that conditions in Corollary 1 hold. Then, the steady state in (16) is a saddle point with two real negative and two real positive eigenvalues, implying a local two-dimensional stable manifold and monotonic convergence to the steady state.

Proof: First, by satisfying the conditions in Corollary 1, we ensure that the steady states in (16) are admissible.

The remaining proof of the proposition proceeds based on the formalism of Dockner and Feichtinger (1991). According to their framework, the steady state defined by equation (16) is a saddle point with a two-dimensional stable manifold if the following conditions are met:

- (a) $|J| > 0$
- (b) $\Phi < 0$
- (c) $\frac{\Phi^2}{4} - |J| \geq 0$,

where $|J|$ denotes the determinant of the Jacobian matrix J .

To verify condition (a), we first compute the Jacobian matrix of the underlying canonical system under the policy $u = u^{\max}$. This matrix is given by

$$J = \begin{bmatrix} \frac{\beta(N-I) - \beta I}{N} - \alpha Q - \gamma & -\alpha I & 0 & 0 \\ -\delta & 0 & 0 & 0 \\ \frac{a_1 N + 2\beta\varphi_I}{N} & \alpha\varphi_I & \rho + \frac{(\alpha Q - \beta + \gamma)N + 2I\beta}{N} & \delta \\ \alpha\varphi_I & a_2 & \alpha I & \rho \end{bmatrix} \quad (18)$$

Evaluating the determinant at the steady state defined by (16) yields

$$|J| = \frac{\alpha u^{\max} z (\alpha u^{\max} z N \delta - N \delta \rho^2 - \beta \rho u^{\max} z)}{\delta N} \quad (19)$$

This expression is positive if and only if $\alpha N \delta - \beta \rho > 0$ and $u^{\max} > \frac{N \delta \rho^2}{z(\alpha N \delta - \beta \rho)}$, which are the conditions stated in Corollary 1 (see 17(i), 17(ii), respectively) and required in the proposition. The magnitude of J , denoted by Φ and defined above, is

$$\Phi = - \frac{u^{\max} z (2 \alpha N^2 \delta^2 + N \beta \delta \rho + \beta^2 u^{\max} z)}{\delta^2 N^2} \quad (20)$$

which is always negative, thereby satisfying condition (b).

Finally, to verify condition (c), we substitute equations (19) and (20) into $\frac{\Phi^2}{4} - |J|$, yielding

$$\frac{\left(N^2 \alpha \delta^2 + \frac{\beta^2 u^{\max} z}{4} \right) z u^{\max} (N \delta \rho + \beta u^{\max} z)^2}{\delta^4 N^4} > 0 \quad (21)$$

which confirms the third condition. Therefore, all three conditions are met, and the steady state defined in (16) is a saddle point with a two-dimensional stable manifold. As a result, the optimal path converges monotonically to the steady state (see Ouardighi et al., 2014).

This concludes the proof. ■

Notably, the conditions stated in Corollary 1, which ensure the admissibility of the steady states in (16), are also sufficient to guarantee that the steady state in (16) is a saddle point with two real negative and two real positive eigenvalues. Additionally, an important insight arises from the condition $u^{\max} > \frac{N \delta \rho^2}{z(\alpha N \delta - \beta \rho)}$, which, if violated, causes the determinant of the Jacobian matrix evaluated at the steady state (see (19)) to become negative. According to Dockner and Feichtinger (1991), this indicates that the system possesses a one-dimensional stable manifold. Consequently, the likelihood that trajectories will converge to the steady state from the initial conditions is reduced. This observation is valuable for decision-makers as it highlights the importance of selecting sufficiently high maximal stock replenishment effort to ensure that the disease reaches and maintains an endemic state in the long run. Failure to meet this requirement could prevent the system from reaching this equilibrium, leading to further disease spread within the population. However, maximal stock replenishment effort $u^{\max}$ is not only bounded from below by $u^{\max} > \frac{N \delta \rho^2}{z(\alpha N \delta - \beta \rho)}$ but also from above by condition 17(ii) stated in Corollary 1. A violation of the upper bound will result in non-admissible steady states.

2.3. Moderate replenishment policy ($0 < u < u^{\max}$)

Under this policy, neither a zero nor a maximum replenishment effort is optimal. We begin the analysis by determining the steady states admitted by the moderate replenishment policy.

Proposition 6. The steady state may not be unique, as it depends on the value of Z^* , which is determined by the quadratic equation

$$AZ^2 + BZ + C = 0, \quad (22)$$

with coefficients

$$A = N^2 \alpha_1 \alpha^2 z^2 + N^2 \alpha_3 \alpha^2 \delta^2 - N \alpha_3 \alpha \beta \delta \rho + a_2 \beta^2 z^2, B = -N \alpha_3 \alpha \delta \rho^2 - a_2 \beta^2 z^2 + a_2 \beta \gamma z^2 + a_2 \beta \rho z^2 \text{ and } C = a_2 \rho z^2 (\gamma - \beta) \quad (23)$$

the corresponding steady-state values are given by

$$I^{ss} = NZ^*, Q^{ss} = \frac{(1 - Z^*)\beta - \gamma}{\alpha}, \varphi_I^{ss} = \frac{(-N \alpha_3 \alpha \delta \rho + a_2 \beta z^2)Z^* + a_2 z^2 (\gamma - \beta)}{N \alpha^2 z^2 Z^*}, \varphi_Q^{ss} = \frac{N \alpha_3 \delta}{z^2} Z^* \quad (24)$$

Proof: Substituting $u = \frac{z \varphi_Q}{a_3}$ into equations (4), (5) and (10)–(11), and setting the derivatives equal to zero, yields (24). ■

Corollary 2. If the quadratic equation in (22) admits real roots, then the steady states of I^{ss} , Q^{ss} and φ_I^{ss} in (24) are admissible (i.e., $I^{ss}, Q^{ss} \geq 0$ and $\varphi_I^{ss} \leq 0$), provided the root satisfies the condition $0 < Z^* \leq 1 - \frac{\gamma}{\beta}$ and condition $\gamma - \beta \geq -\frac{(-N \alpha_3 \alpha \delta \rho + a_2 \beta z^2)Z^*}{a_2 z^2}$ holds.

Proof: Suppose the quadratic equation in (22) yields a real and positive root $Z^* > 0$. Then, from (24), it follows that $I^{ss} = NZ^* > 0$. To ensure that the expression for Q^{ss} in (24) is non-negative, we require that $Z^* \leq 1 - \frac{\gamma}{\beta}$. Combining the two inequalities yields the condition $0 < Z^* \leq 1 - \frac{\gamma}{\beta}$ stated in the corollary. Finally, to ensure that $\varphi_1^{ss} \leq 0$, it is sufficient to require that $\gamma - \beta \geq -\frac{(-Na_3\alpha\delta\rho + a_2\beta z^2)Z^*}{a_2 z^2}$ is met. ■

To proceed with the stability analysis of the steady states defined in (24), we consider each admissible solution of the quadratic equation (22) that satisfies the conditions stated in Corollary 2. Specifically, let Z_1 denote any root of (22) such that $0 < Z_1 \leq 1 - \frac{\gamma}{\beta}$ for which the corresponding steady-state values are then computed. Let the steady state corresponding to this root, as defined by (24), be represented by $I_1^{ss}, Q_1^{ss}, \varphi_{I1}^{ss}$ and φ_{Q1}^{ss} . Finally, let J denote the Jacobian matrix of the canonical system in equations (4), (5) and (10)–(11), under the policy $u = \frac{z\varphi_0}{a_3}$, and let Φ denote the magnitude of J . To facilitate the proof, denote $\mathcal{R}_1 = (-\beta^2 Z_1^2 + (-2N\alpha\delta - \rho\beta)Z_1)a_3 - a_2 z^2$, $\mathcal{R}_2 = (N^2\alpha^2 a_1 + a_2\beta^2)z^2 Z_1^2$ and $\mathcal{R}_3 = Na\delta a_3 Z_1^2 (N\alpha\delta - \rho\beta)$.

Proposition 7. Suppose that conditions in Corollary 2 are met. If $0 < u < u^{\max}$, $N > \frac{\beta\rho}{\alpha\delta}$ and $\gamma - \beta > -\frac{\frac{\mathcal{R}_1}{4a_3} - \mathcal{R}_2 - \mathcal{R}_3}{z^2 \rho a_2}$, then the steady state $I_1^{ss}, Q_1^{ss}, \varphi_{I1}^{ss}$ and φ_{Q1}^{ss} , defined by (24), is a saddle point with two real negative and two real positive eigenvalues, implying a local two-dimensional stable manifold and monotonic convergence to the steady state.

Proof: The proof follows a similar structure to the proof of Proposition 5. To show that the steady state is a saddle point with a local two-dimensional stable manifold, it is sufficient, according to Dockner and Feichtinger (1991), to prove that all three conditions stated in the proof of Proposition 5 are met. We begin by verifying the first of these conditions, which requires that $|J| > 0$ in the steady state. To this end, we compute the Jacobian matrix of the canonical system in equations (4), (5) and (10)–(11) under the policy $u = \frac{z\varphi_0}{a_3}$. That is,

$$J = \begin{bmatrix} \frac{\beta(N-I) - \beta I}{N} - \alpha Q - \gamma & -\alpha I & 0 & 0 \\ -\delta & 0 & 0 & \frac{z^2}{a_3} \\ \frac{a_1 N + 2\beta\varphi_I}{N} & \alpha\varphi_I & \rho + \frac{(\alpha Q - \beta + \gamma)N + 2I\beta}{N} & \delta \\ \alpha\varphi_I & a_2 & \alpha I & \rho \end{bmatrix}, \quad (25)$$

and its determinant, evaluated at the steady state $I_1^{ss}, Q_1^{ss}, \varphi_{I1}^{ss}$ and φ_{Q1}^{ss} , is

$$|J| = \frac{\mathcal{R}_2 + \mathcal{R}_3 - z^2 \rho a_2 (\gamma - \beta)}{a_2} \quad (26)$$

Given that $\gamma < \beta$, it is sufficient to require that $N > \frac{\beta\rho}{\alpha\delta}$ in $\mathcal{R}_3$, as stated in the proposition, in order to ensure that $|J| > 0$.

Next, we verify the second condition stated in the proof of Proposition 5, $\Phi < 0$, where Φ is computed as the sum of the principal second-order minors of J (in (25)) minus the squared discount rate. Evaluated at the steady state, Φ takes the form

$$\Phi = \frac{\mathcal{R}_1}{a_3} \quad (27)$$

which is always negative, and thus the second criterion is satisfied.

Finally, to verify the third condition, substituting (26) and (27) into $\frac{\Phi^2}{4} - |J|$ yields

$$\frac{\mathcal{R}_1^2}{a_3^2} - \frac{4(\mathcal{R}_2 + \mathcal{R}_3 - z^2 \rho a_2 (\gamma - \beta))}{a_3}, \quad (28)$$

which is positive if

$$\gamma - \beta > -\frac{\frac{\mathcal{R}_1^2}{4a_3} - \mathcal{R}_2 - \mathcal{R}_3}{z^2 \rho a_2} \quad (29)$$

as stated in the proposition.

This concludes the proof. ■

This section has presented the theoretical results concerning the steady states, their properties, and the associated transition paths under three replenishment regimes: zero, moderate, and maximum (as defined by the control variable). To

support these analytical findings and to further investigate the system's long-run behavior, in the following section, we present a numerical analysis in which we vary selected parameters to explore the sensitivity of the system dynamics.

3. Computational experiments

We conduct an extensive set of numerical experiments based on the parameter values provided in Table 1. Note that the disease-related values in Tables 1 i.e., transmission rate (β) and baseline recovery rate (γ), are derived from the epidemiology of gonorrhoea. In particular, in a frequency-dependent transmission framework, the transmission parameter β can be interpreted as a product of the effective contact rate and the per-contact transmission probability. The literature reports that the per-contact transmission probability may vary by the direction of transmission; from male to female, it stands in the range 0.2–0.3 (see Van Duynhoven, 1999). Assuming one effective contact per model time unit, we select $\beta = 0.3 \left[\frac{1}{\text{time unit}} \right]$ as a representative value corresponding to a scenario of relatively efficient transmission of gonorrhoea within the population. In turn, γ is the inverse of the average duration time of the infection. Empirical studies (see, for instance, Kiss et al., 2024) report spontaneous clearance of gonorrhoea typically occurring within approximately one to two weeks. Interpreting the model time unit accordingly, we take $\gamma = 0.1 \left[\frac{1}{\text{time unit}} \right]$, corresponding to a mean infectious period of approximately 10 model time units, which is in line with the described clearance times. While these values are based on published studies and are biologically reasonable, we note that the data in Table 1 are intended for illustrative purposes. The obtained numerical results provide qualitative insights into the model's dynamics rather than precise clinical predictions.

The selection of gonorrhoea as the testbed disease is not arbitrary. Despite the availability of protective measures and public health campaigns, global infections reached 82.4 million among adults in 2020 (World Health Organization, 2024). Although the disease is treatable with antibiotics, infected individuals do not develop lasting immunity and can be reinfected after subsequent exposure. Moreover, if left untreated, gonorrhoea may lead to serious medical consequences, including infertility (in men) and blindness in infants born to infected mothers (Cleveland Clinic, 2024). Early treatment significantly reduces the risk of adverse health consequences, making access to antibiotics an essential factor in disease management.

The parameter values in Table 1 may vary, and such changes can affect both the system's steady states and its dynamic behavior. Accordingly, we conduct a sensitivity analysis to investigate the impact of these parameters under both the maximal (Case 1) and the moderate (Case 2) replenishment policy. The analysis focuses on a subset of influential parameters to ensure clarity and avoid excessive complexity. The structure of the sensitivity analysis is hierarchical. First, we present the main, steady-state findings for Cases 1 and 2. Then, we examine the findings of each analysis in more detail, including the dynamic behavior, to extract relevant managerial insights.

Also note that although the optimal control problem is formulated on an infinite time horizon, numerical simulations are conducted on a finite interval. The interval is chosen to be long enough (e.g., $T = 40$ for Case 1 and $T = 130$ for Case 2), to ensure that all trajectories reach a steady state well before the end of the simulation period.

3.1. Main results for maximum and moderate replenishment: steady states

In each investigated scenario, a single parameter is varied while all others are held constant. The analysis focuses on the impact of three key parameters: β , γ , and z . Biologically, β captures the disease speed, while γ reflects the natural recovery of the infected individuals. In turn, z represents a constraint on how rapidly treatment availability can be increased in the population per replenishment cycle. Plotting the dynamics for a broad range of β , γ , and z not only validates the theoretical results of stability but also provides epidemiological insights into the impact of these parameters on the infection levels and the time necessary for the stabilization under the various controlled interventions.

Table 2
Sensitivity analysis results – Case 1.

Scenario (#1)	I^0 (#2)	Q^0 (#3)	I^{ss} (#4)	Q^{ss} (#5)
β^*				
0.3	51	171.78	53.33	184.00
0.4	51	266.56	53.33	278.66
0.45	51	313.95	53.33	326.00
γ				
0.01	51	261.78	53.33	274.00
0.05	51	221.78	53.33	234.00
0.1	51	171.781	53.33	184.00
z				
5	20	138.11	26.66	192.00
7	30	139.98	37.33	188.80
10	51	171.78	53.33	184.00

Table 3
Sensitivity analysis results - Case 2.

Scenario (#1)	I^0 (#2)	Q^0 (#3)	$u(0)$ (#4)	I^{ss} (#5)	Q^{ss} (#6)	u^{ss} (#7)
β^*						
0.15 ^a	15	20	6.97	24.73	46.28	3.71
0.25	34	120	1.9	45.76	138.55	6.86
0.27	40	140	4.6	49.40	156.65	7.41
γ						
0.05	32	120	0.72	45.25	140.94	6.78
0.1	37	103	0.57	36.14	92.77	5.42
0.15	15	20	6.57	24.68	45.06	3.7
z						
7	25	78	1.18	35.84	92.83	7.68
14	30	80	3.31	36.28	92.74	3.88
18	30	78	3.67	36.34	92.73	3.02

^a The baseline scenario for Case 1 assumes that $\beta = 0.3$, whereas for Case 2, the baseline scenario sets $\beta = 0.2$.

The findings for Case 1 (maximum replenishment policy) are summarized in Table 2 and illustrated in Figs. 1–4, while the results for Case 2 (moderate replenishment policy) are summarized in Table 3 and depicted in Figs. 5–9.

The structure of Tables 2 and 3 is broadly similar, except that in Table 3, along with the initial and steady-state values of the state variables, we also report the initial and terminal values of the control variable. Specifically, Columns 2–3 in both tables report the initial values of the state variables, while the corresponding steady-state values are shown in Columns 4–5

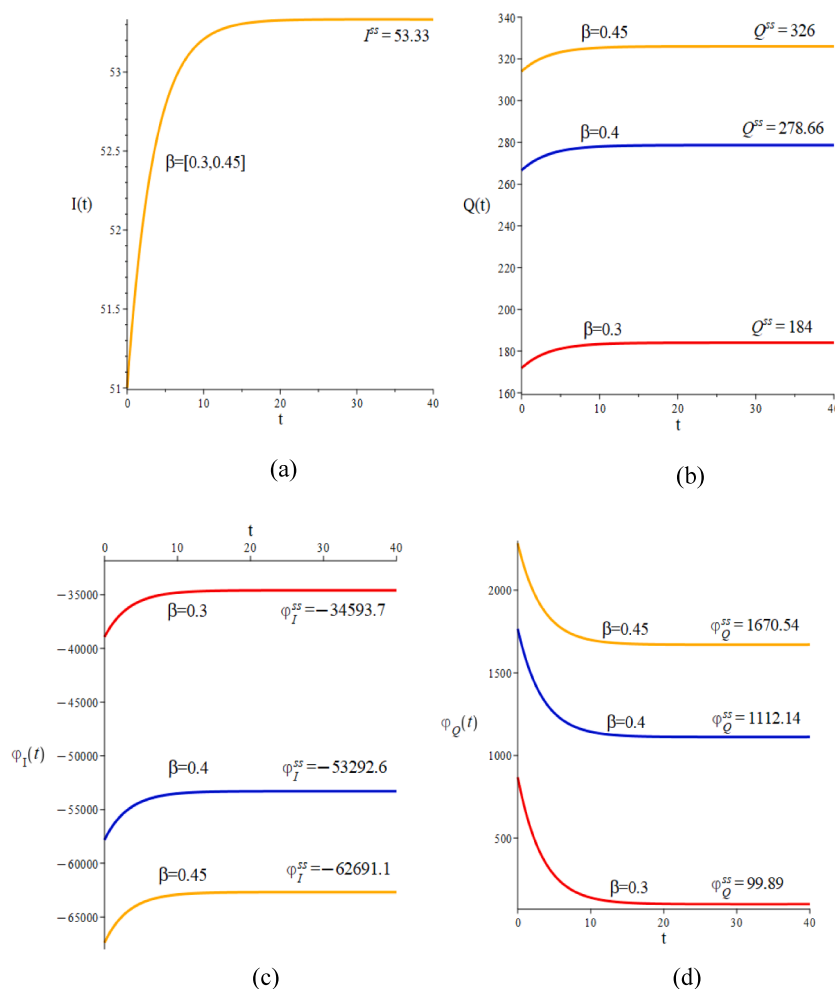


Fig. 1. The evolution of (a)–(b) the state variables and (c)–(d) the co-state variables over an infinite planning horizon for different values of β

of Tables 2 and in Columns 5–6 of Table 3. Further, for the moderate replenishment policy (Table 3), Columns 4 and 7 respectively show the initial and final value of the control variable.

An important observation from Tables 2 and 3 is that the appropriate choice of replenishment policy (either maximum or moderate) depends on the underlying parameter values. Specifically, when the disease transmission rate, β , is relatively low, it is reasonable to adopt the moderate replenishment regime (see Table 3). However, as β increases, this policy becomes inadequate, and the maximum replenishment policy (see Table 2) is preferred instead. Additionally, as the maximum delivery quantity per replenishment, z , increases, it becomes profitable to activate the moderate replenishment regime (Case 2). This observation reflects a natural trade-off, where a larger delivery quantity can compensate for a slower replenishment rate, and vice versa.

The following sections provide a detailed analysis of the impact of the investigated parameters on the steady states of the state and co-state variables, as well as on the behavior of the optimal saddle-path trajectories.

3.2. Steady states and optimal saddle paths: Maximum replenishment (Case 1)

To explicitly link our numerical results with the theoretical framework, we present the dynamics for Case 1 in Figs. 1–3. Specifically, for the data set presented in Table 1, conditions in Corollary 1 are satisfied, which ensures that the endemic steady state, defined by equation (16) in Proposition 4, is admissible. Furthermore, since the conditions of Corollary 1 hold, Proposition 5 guarantees that the convergence to the state is monotonic. The numerical simulations align with this theory: the saddle paths in Figs. 1–3 demonstrate expected monotonic convergence to the endemic equilibrium.

A key insight is that, regardless of the input parameters, the system converges to the steady state relatively quickly when the maximum replenishment policy is applied. This finding has practical implications for the decision-maker: if the primary goal of the healthcare system is to stabilize the situation as rapidly as possible, adopting the maximum replenishment regime is a justified strategy—while it may incur higher costs, it ensures relatively fast convergence of the system to the

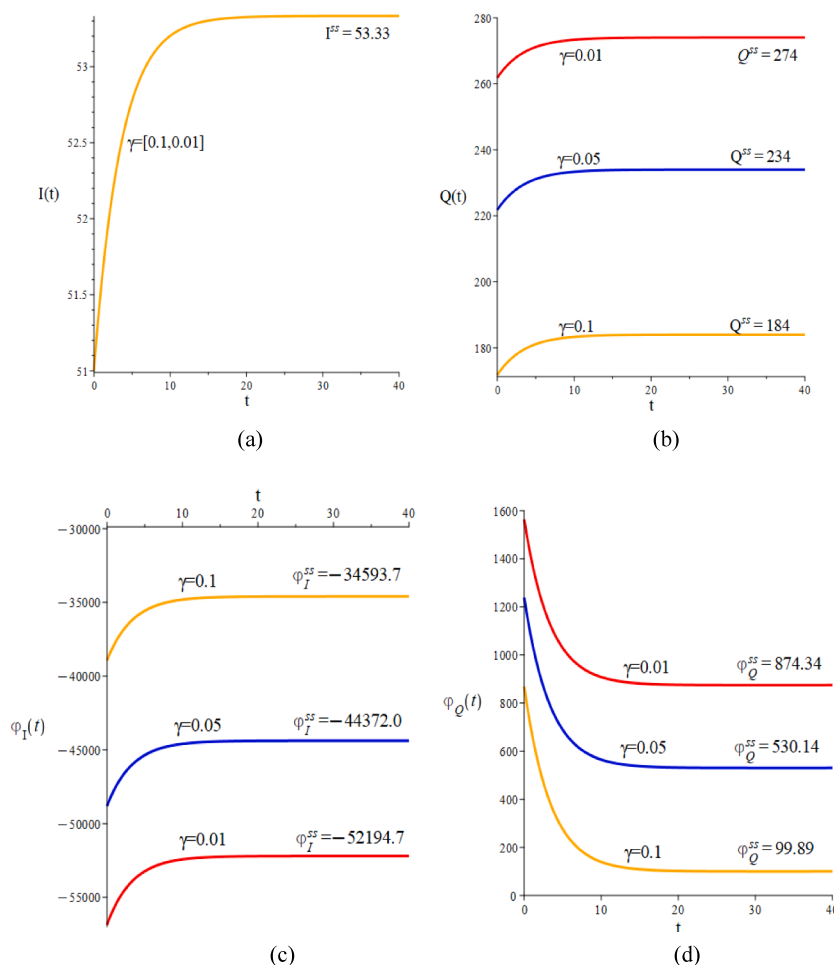


Fig. 2. The evolution of (a)–(b) the state variables and (c)–(d) the co-state variables over an infinite planning horizon for different values of γ

steady state. The dynamics of the control variable u are not displayed in these figures, as $u = u^{max}$ remains unchanged throughout the entire time horizon.

The first sensitivity analysis focuses on varying the disease transmission rate, β , and is presented in Fig. 1.

Fig. 1(a) shows monotonic convergence of the disease dynamics toward a unique endemic steady state I^{ss} , verifying the theoretical results established in Propositions 4–5. However, Fig. 1(a) reveals a surprising result: the entire trajectory of the infected population size, from the initial time point to steady state, remains unchanged regardless of the value of β . This finding may be of practical value for healthcare planners: it simplifies resource allocation and planning, as no unexpected fluctuations or surges in infection rate are anticipated over the planning horizon. In contrast to I , the steady-state value of the inventory Q is sensitive to changes in β . As β increases, Q^{ss} also increases (see Fig. 1(b)). This result is intuitive: a higher disease transmission rate requires a larger stock of medication to manage the increased demand and control disease spread. Next, we examine the impact of the baseline recovery rate, γ , as shown in Fig. 2.

As was the case for β (see Fig. 1(a)), the baseline recovery rate γ does not influence the steady-state value of the infected population, I^{ss} , (see (16)), as shown in Fig. 2(a). Additionally, the trajectory of the number of infected individuals remains unchanged regardless of γ . However, as γ increases, the steady-state value of the medicine stock, Q^{ss} , decreases (see Fig. 2(b)). This outcome is intuitive: a higher natural recovery rate reduces the need for external medical intervention, allowing Q^{ss} to stabilize at a lower level. The influence of the maximum delivery quantity per replenishment (z) is analyzed next, in Fig. 3.

In contrast to β and γ , which do not affect the steady-state value I^{ss} , the maximum delivery quantity per replenishment, z , does have an influence. As z increases, a greater number of infected individuals persist at the steady state (see Fig. 3(a)). The effect of z on the steady-state medicine stock, Q^{ss} , is quite different. As z decreases, Q^{ss} increases (see Fig. 3(b)), while the initial inventory level, Q^0 , becomes smaller. Consequently, when z is low, the system takes longer to reach a steady state than

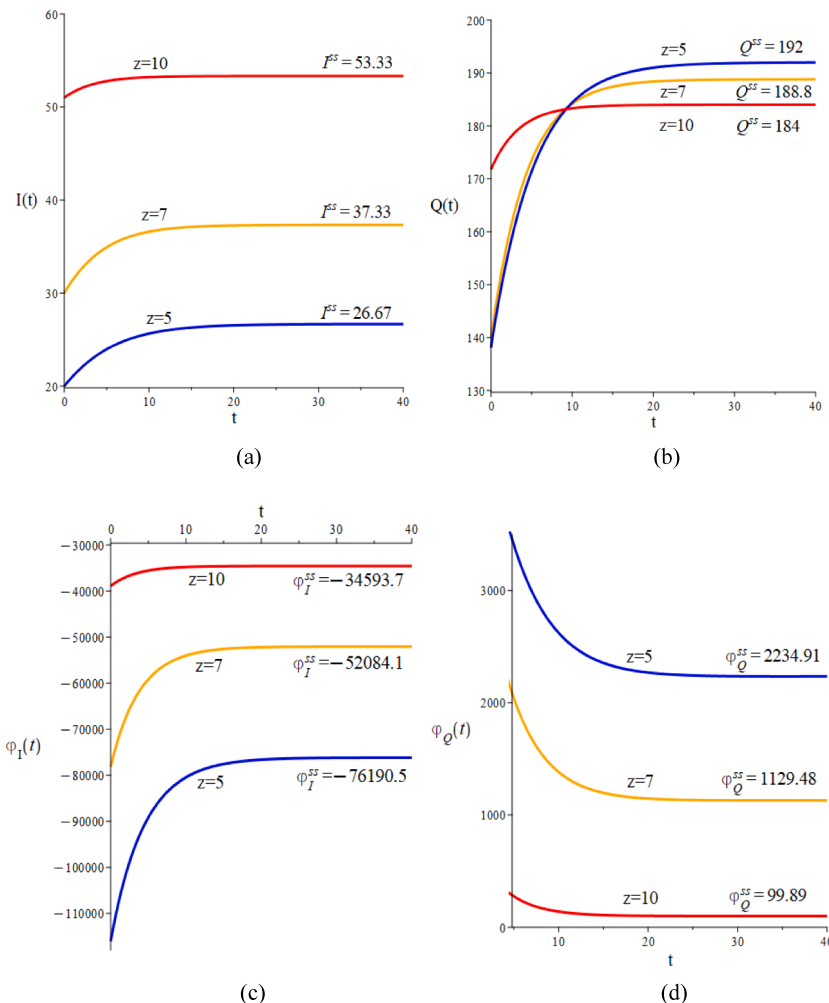


Fig. 3. The evolution of (a)–(b) the state variables and (c)–(d) the co-state variables over an infinite planning horizon for different values of z

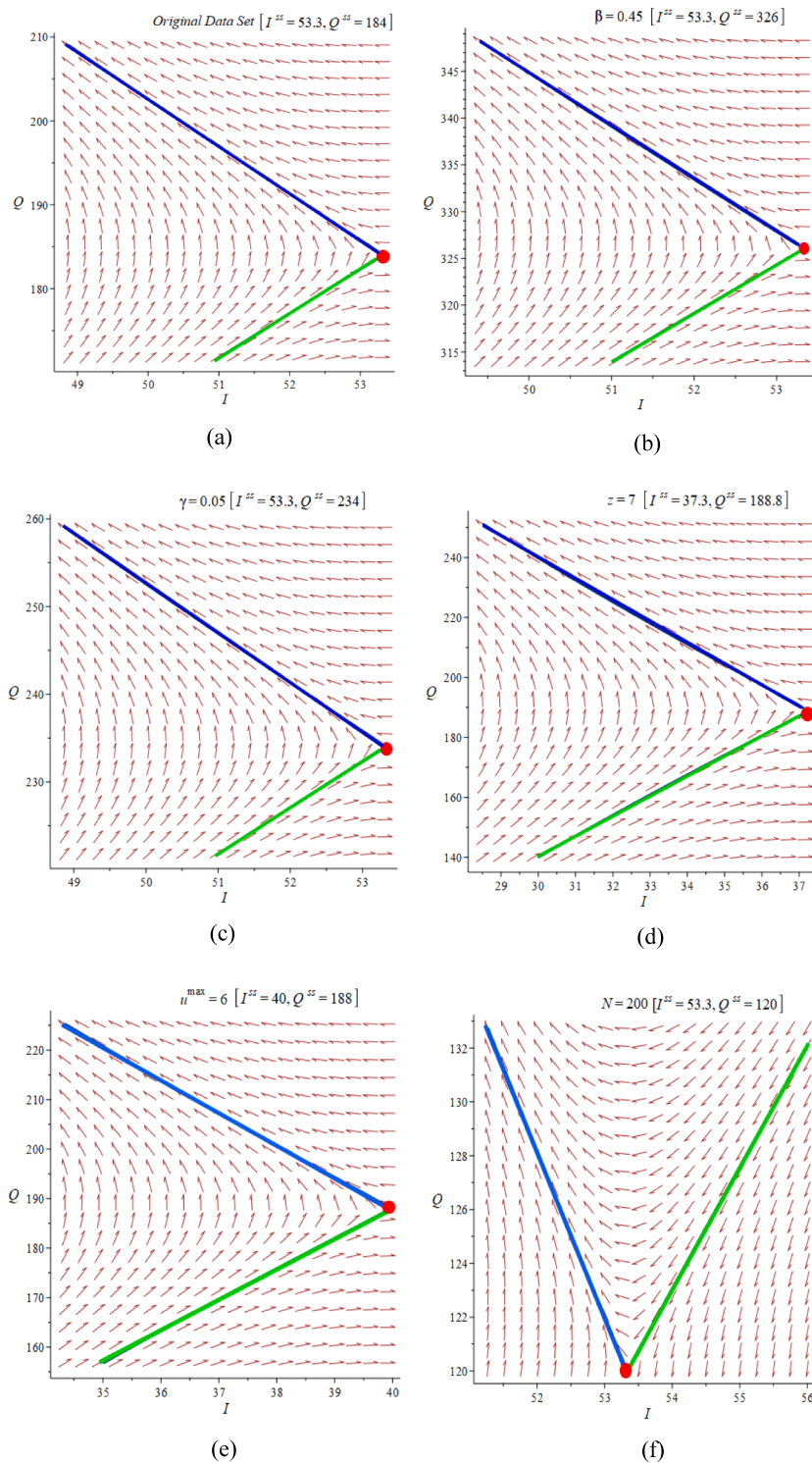


Fig. 4. Stable and unstable manifolds in $I - Q$ space under: (a) the baseline setting (see Table 1), (b) $\beta = 0.45$, (c) $\gamma = 0.05$, (d) $z = 7$, (e) $u^{max} = 6$, and (f) $N = 200$

in scenarios with a higher delivery capacity. This suggests that, when logistics limitations prevent a large delivery capacity per replenishment cycle, healthcare planners should anticipate a longer stabilization period.

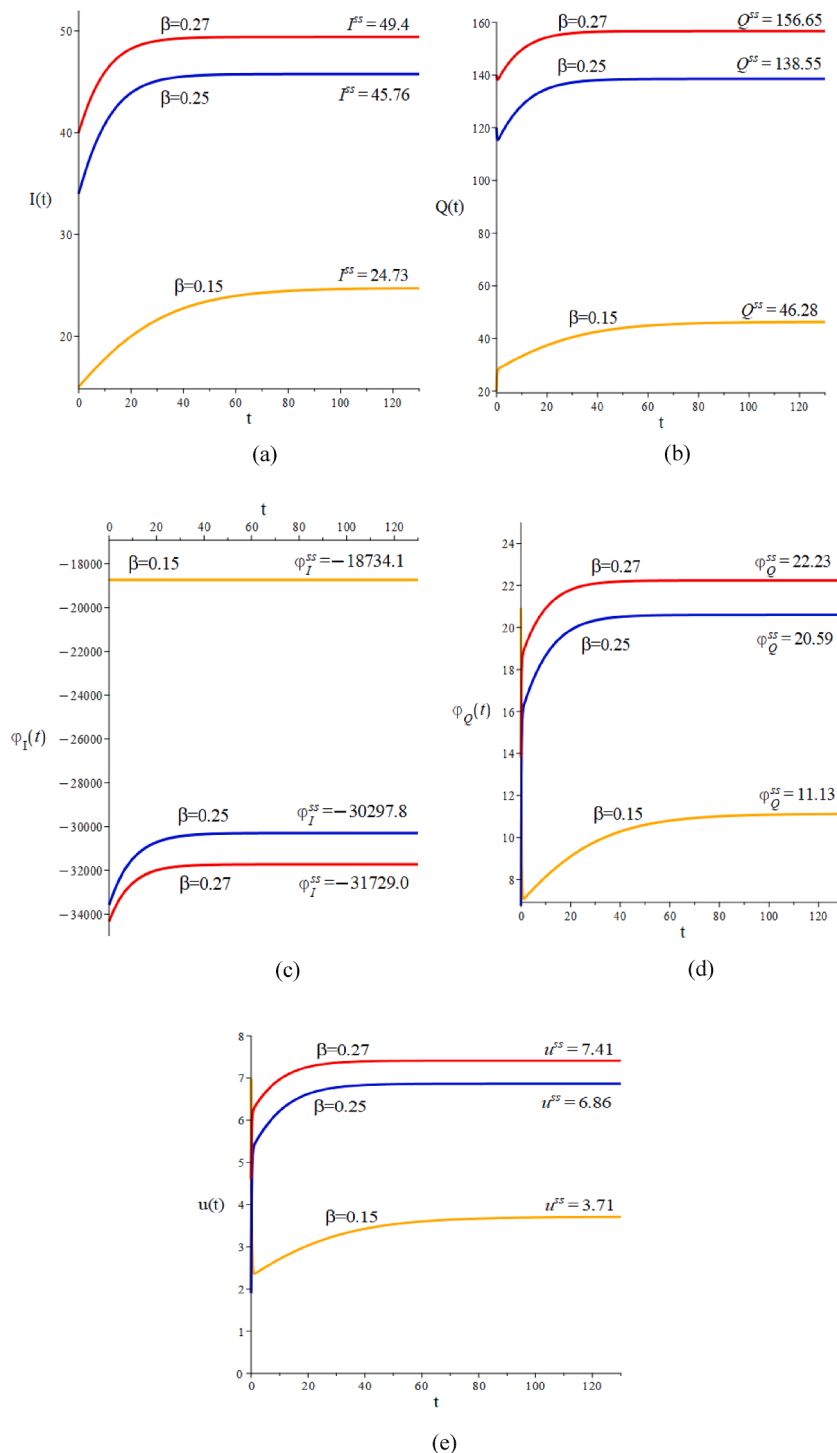


Fig. 5. Evolution of (a)–(b) the state variables, (c)–(d) the co-state variables, and (e) the control variable over an infinite planning horizon for different values of β

The above numerical results provide not only logistics-oriented insights but also clear biological interpretations. Specifically, the data demonstrates that under maximal replenishment effort, the disease is steered toward a predictable, controlled endemic state for the broad range of β, γ , and z values. Biologically, this indicates that the intervention consistently controls disease spread despite variations in the model's parameters. Monotone convergence means that infections move steadily toward the steady state, without ups and downs or repeated waves, providing a stable and manageable public

health environment. Furthermore, from a biological perspective, the invariance of the infection trajectory shown in Figs. 1(a) and 2(a) indicates that under maximal control, the intervention governs the disease behavior and effectively limits the impact of fluctuations in disease severity or natural recovery.

Finally, we present the phase portrait of the system under maximum replenishment efforts for various parameter settings (see Fig. 4). Specifically, in Fig. 4, the saddle steady state is designated by the red point. Green and blue colours mark the stable and unstable manifolds of the saddle steady state, respectively. The values of the steady states are mentioned in the squared brackets near the caption. A common feature across Fig. 4(a)–(f) is that along the stable manifold (see green-marked line), a higher initial level of the infected population I^0 must be accompanied by higher initial stocks of medication Q^0 in order for the system to converge to the saddle steady state. This relationship can be interpreted as the “cost” incurred when an additional infected individual enters the system, in terms of the increased stock required.

3.3. Steady states and optimal saddle paths: Moderate replenishment (Case 2)

The dynamics of the infection under the moderate replenishment regime are presented in Figs. 5–8. These numerical simulations confirm the analytical results reported in Corollary 2 and Propositions 6–7. For the given data set, the conditions of Corollary 2 are satisfied, implying that the steady-state defined in equation (24) of Proposition 6 is admissible. Under this parametrization, the equilibrium corresponds to an endemic steady state. Moreover, since the population size satisfies $N = 1,000 > \frac{\beta I^0}{\alpha \delta}$ and the net infection growth rate $\beta - \gamma$ is bounded from above by the condition stated in Proposition 7, the infection dynamics exhibit monotonic convergence toward this steady state. Once reached, the disease remains at the equilibrium over the entire numerical simulation horizon.

In comparison with Case 1, the system under Case 2 takes significantly longer to stabilize. This outcome is expected: with a lower replenishment effort, the system exerts less control over the disease dynamics, resulting in slower convergence. As in Case 1, we investigate the sensitivity of the system to changes in β (Fig. 5), γ (Fig. 7), and z (Fig. 8). In addition to the dynamics of the state and co-state variables, which were also reported for Case 1, Figs. 5, 7 and 8 present the evolution of the control variable over time.

In contrast to Case 1, where β has no impact on I^{ss} (Fig. 1(a)), under Case 2, an increase in β leads to a higher value of I^{ss} . This increase in I^{ss} , in turn, leads to an increase in Q^{ss} , as a larger inventory is needed to treat the larger infected population. One of the most striking findings concerns the dynamics of the control variable, shown in Fig. 5(e). When β is large, i.e., when the disease spreads rapidly, the control variable rises very sharply from its initial value, almost reaching the steady-state value within a short time period. For example, for $\beta = 0.25$, the initial replenishment rate is $u(0) = 1.9$ (see Table 3), but within a very short time, the control trajectory approaches the steady-state value. Thereafter, the trajectory converges toward the steady state more slowly. This behavior can be explained as follows: a higher transmission rate causes a disease outbreak, which can be suppressed by immediately increasing the replenishment frequency. Then, once adequate control has been achieved, the system gradually converges toward the steady state. For decision-makers, this implies that the logistics capacity required for a rapid response must be available immediately upon request, as any delay in mobilizing resources (e.g., due to lack of availability of delivery trucks) could result in a situation where the disease enters the epidemic phase, and the stable endemic equilibrium will be missed. Conversely, when β is low, the system may start with a high replenishment rate (e.g., for $\beta = 0.15$; $u(0) = 6.97$; see Table 3), but the demand for replenishment decreases rapidly thereafter (see Fig. 5(e)).

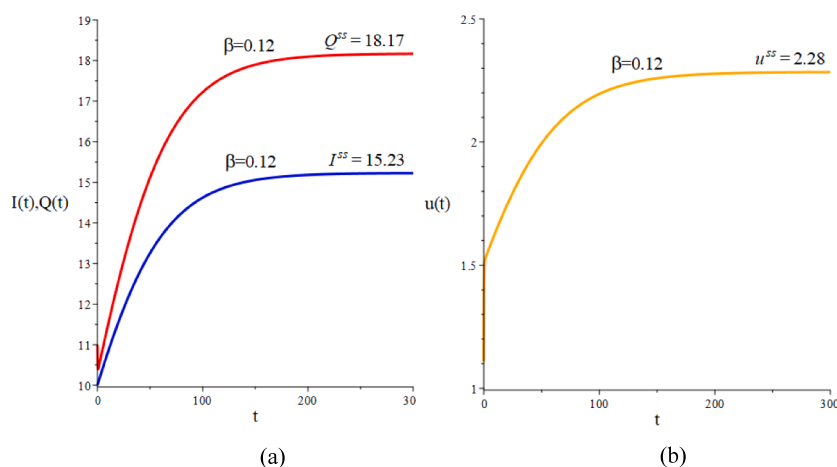


Fig. 6. The evolution of (a) the state variables and (b) the control variable when $\beta = 0.12$

From Fig. 5, it follows that the system in Case 2 requires more time to reach stability compared to Case 1. However, as β continues to decrease—for example, to 0.12—the time needed to reach the steady state increases substantially (see Fig. 6(a) and (b)). This result may appear counterintuitive: one might expect that a lower transmission rate would lead to faster stabilization. The opposite occurs because, as shown in Fig. 5(e), a decrease in β reduces the replenishment efforts, which in turn prolongs the time needed for the system to stabilize.

Fig. 7 illustrates the behavior of the optimal saddle-path trajectories for the state, co-state, and control variables under different values of the recovery rate γ . These trajectories resemble those in Fig. 4, but with opposite trends. For example, whereas increasing the transmission rate β leads to higher steady-state infection levels (I^{ss}) in Fig. 5(a), in Fig. 7(a), the infection levels increase as γ decreases. This inverse relationship is expected: when the baseline recovery rate is higher, infected individuals recover more quickly under the same control strategy, leading to a lower steady-state infection level.

Finally, Fig. 8 depicts the convergence dynamics of the state, co-state, and control variables over an infinite planning horizon for different values of z .

Fig. 8 shows that as the delivery capacity per replenishment cycle (z) increases, the required replenishment effort decreases (see Fig. 8(e)). This outcome is intuitive. However, a less intuitive outcome is observed when z is small—despite the limited delivery capacity, the initial replenishment effort, $u(0)$, is very low (see Table 3). To compensate for the limited delivery capacity, the control effort increases substantially over a very short time period and quickly approaches its steady-

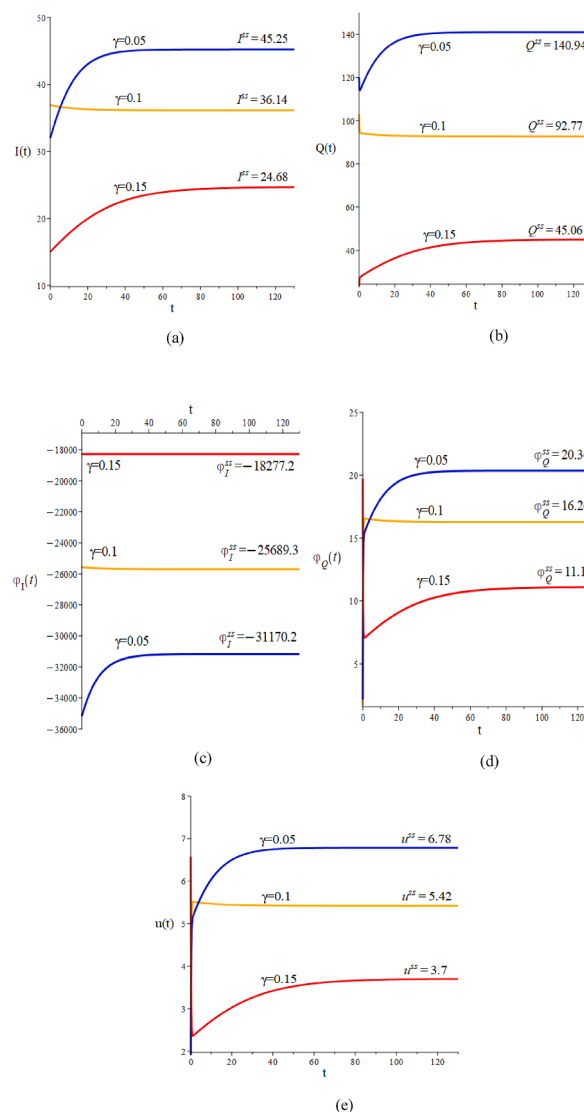


Fig. 7. The evolution of (a)–(b) the state variables, (c)–(d) the co-state variables, and (e) the control variable over an infinite planning horizon for different values of γ .

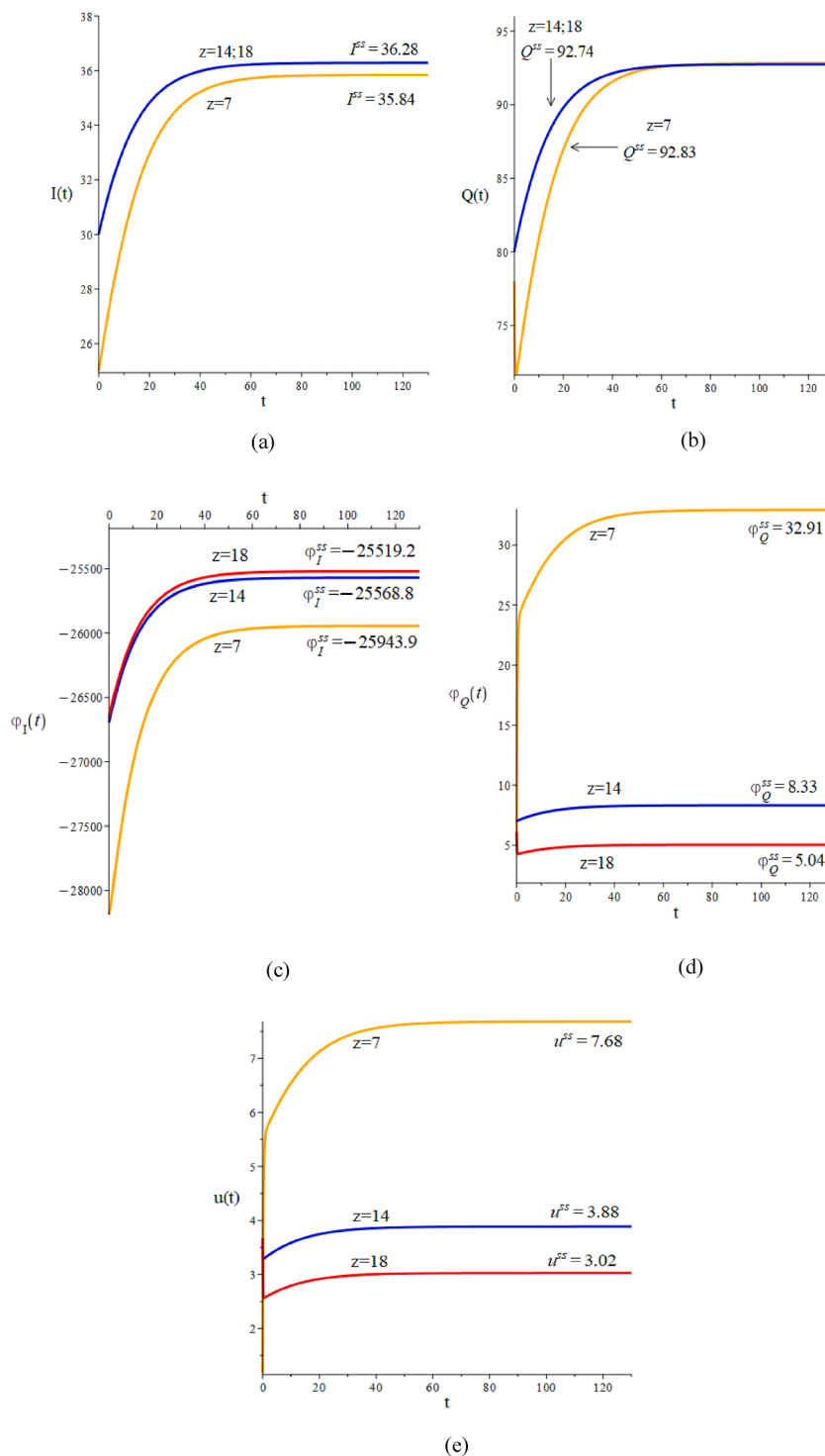


Fig. 8. The evolution of (a)-(b) the state variables, (c)-(d) the co-state variables, and (e) the control variable over an infinite planning horizon for different values of z

state value. In contrast, when z is larger, the initial control effort starts at a higher level and then rapidly declines because the ability to deliver a larger quantity per cycle reduces the need for frequent replenishment.

Another noteworthy result arises from Fig. 8(a). As z increases, the steady-state level of infection, I^{ss} , also rises. However, beyond a certain point, further increases in z no longer affect I^{ss} , suggesting the existence of an upper bound. Thus, it appears

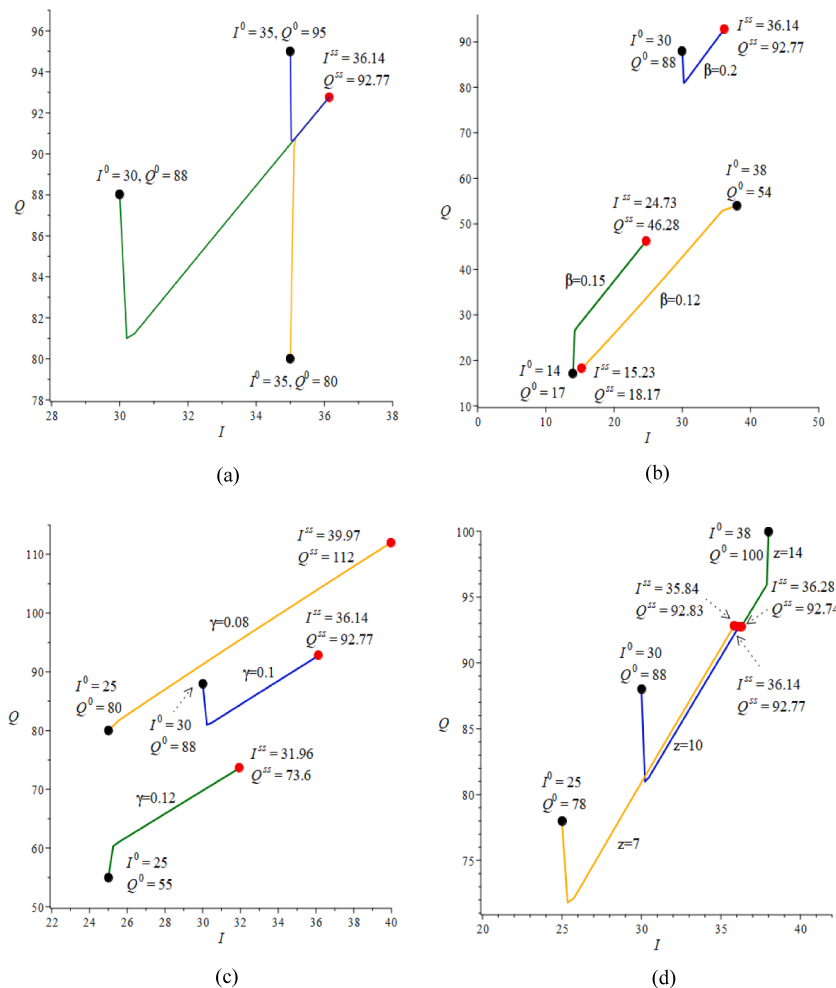


Fig. 9. Projected trajectories in $I - Q$ space under: (a) the baseline setting (see Table 1), and alternative parameter configurations for (b) β , (c) γ , and (d) z

that the system naturally limits the number of infected individuals at steady state, even when the delivery capacity z continues to increase. This result may be valuable for decision-makers, as it implies that only a finite amount of personnel and other resources are required in the long run. Once these needs are met, additional capacity can be redirected to other areas within the healthcare system that more urgently need attention.

From the biological perspective, similar to *Case 1*, under a moderate replenishment rate, the disease monotonically converges to an endemic steady state. It remains at this equilibrium within the simulated time horizon. However, a moderate replenishment rate imposes a longer time necessary for the disease to stabilize and achieve a constant level. Biologically, this indicates a prolonged transient phase, which increases the cumulative disease burden before stabilization.

Similarly to *Case 1*, to illustrate the system's dynamics, we employ phase-diagram type visualizations. However, unlike in *Case 1*, where under maximum replenishment efforts the state variables $I(t)$ and $Q(t)$ form a decoupled two-dimensional system and a classical phase portrait can be constructed, the situation in *Case 2* is quite different. In particular, under moderate replenishment effort, the control variable depends explicitly on the co-state variable φ_Q , namely $u = \frac{z\varphi_Q}{a_3}$, which results in a four-dimensional state-co-state system. Accordingly, a complete phase portrait with a global vector field is not well defined in the (I, Q) plane. Hence, Fig. 9 presents projected trajectories that provide solution paths and convergence properties of the system for various data settings, rather than a classical phase portrait. Note that the black points in Fig. 9(a)–(d) denote the initial state of the system, while the red points indicate the corresponding steady state. We observe that, regardless of the explored initial conditions, the trajectories converge to the steady state. However, the time required for convergence depends on both the initial state and the parameter configuration. For instance, a decrease in the disease transmission rate β leads to a longer convergence time, as illustrated in Fig. 9(b). These results align with the numerical simulations reported in Fig. 6.

4. Conclusion

Non-immunizing diseases, which do not grant long-term immunity after recovery, can be a burden to the economy and can pose significant challenges for healthcare planning. Because such diseases cannot be eradicated, healthcare providers need to maintain continuous vigilance and surveillance to avoid disease escalation. In such cases, guiding the disease into a stable endemic state can be an effective long-term strategy: although the disease will persist for a long time, large-scale outbreaks will be avoided.

In this study, we use the classical SIS epidemic model to investigate disease dynamics over a long horizon. In reality, disease progression depends on a variety of exogenous and endogenous factors. We suggest taking account of logistical factors; specifically, we extend the standard SIS model by introducing a new dynamic equation to represent the inventory of available medication. This inventory is managed via the replenishment rate, and it is linked to disease dynamics through the recovery rate: the larger the available medical stock, the faster the recovery of infected individuals and the sooner they return to the susceptible class.

To avoid situations where frequent adjustments to replenishment rates are required during the planning period, we focus on three pure strategies: zero, moderate, and maximum replenishment effort. Our key findings are as follows: Under the zero-replenishment regime, the steady state cannot be attained by any admissible trajectory, and the infection persists in the population. From a managerial view, this indicates that implementing the zero-replenishment control alone is not enough to contain disease spread. Accordingly, this regime should be used in combination with other proactive interventions that are directed toward reducing the disease burden (e.g., screening initiatives and early detection programs). The steady state is unique under the maximum replenishment rate, but it may not be unique under moderate replenishment. The steady states under maximum and moderate replenishment efforts are saddle points—with a local two-dimensional stable manifold—and convergence to the steady state is monotonic, provided that the conditions stated in [Propositions 5 and 7](#) are satisfied. Finally, we find that under the maximum replenishment policy, the stability of the steady state depends on the maximum replenishment rate. If this rate falls below a certain threshold, the system's stability is compromised.

Our numerical experiments validate the reported theoretical results by demonstrating that, for the given dataset, the infection dynamics under both maximum and moderate replenishment efforts converge monotonically to the endemic equilibrium and remain there throughout the simulated time horizon. Moreover, they reveal several practical insights. First, when the disease transmission rate, β , is relatively low, it is appropriate to adopt the moderate replenishment regime. Second, the system converges to the steady state faster under the maximum replenishment policy than under the moderate replenishment policy. Third, we find that under the maximum replenishment policy, the optimal saddle-path trajectories of the infected population are independent of the disease transmission rate (β) and the baseline recovery rate (γ). This result could be valuable for healthcare planners because it allows them to plan resource use more easily, knowing that sudden spikes or drops in the number of infected people during the planning period are not expected. Biologically, this means that under maximum replenishment effort, system behaviour is determined by the intervention that effectively restricts the influence of fluctuations in β and γ on the infection dynamics. When the delivery capacity per replenishment cycle (z) decreases, the convergence to the steady-state inventory level, Q^{ss} , slows. This suggests that under constrained logistics, where disease stabilization might be delayed, healthcare planners may need to account for longer stabilization periods.

Under the moderate replenishment policy, when the disease spreads rapidly (high β), immediate and sharp increases in the stock replenishment rate are crucial to avoid losing control of the outbreak. This highlights the importance of having logistics resources ready for rapid deployment at the beginning of the planning period—to avoid the situation where the system fails to reach the endemic steady state and instead escalates into an epidemic. Finally, we observe that as the maximum delivery quantity per replenishment cycle (z) increases, the steady-state number of infected individuals, I^{ss} , only increases up to a finite limit. This insight is valuable for decision-makers, as it imposes a cap on long-term resource requirements, allowing excess resource capacity to be redirected to other critical areas.

This study makes a novel contribution by linking long-term disease dynamics with inventory management considerations. Nonetheless, it has a number of limitations that suggest promising avenues for future research. First, the deterministic setting does not fully capture the uncertainties of real-world disease spread; thus, extending the model to a stochastic inventory-epidemic framework is a promising direction. Second, the assumption of instantaneous deliveries is a simplification; incorporating delayed optimal control into the model would help address this limitation and improve its alignment with real-world logistics.

Data availability statement

All data used in this study are entirely included within this article.

Funding

The authors did not receive support from any organization for the submitted work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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